Louis Arts

London, UK | artslouis@gmail.com | LinkedIn | GitHub | Personal Website

Education

University College London — London, UK
MSc Data Science and Machine Learning, Distinction — *Sep 2025 – Sep 2026*

Imperial College London — London, UK
MSc Physics, Merit — *Sep 2023 – Sep 2024*

Le Wagon — Zurich, Switzerland
Data Science Bootcamp — *Oct 2022 – Dec 2022*

Maastricht University — Maastricht, Netherlands
BSc Liberal Arts and Sciences (Physics), First-Class Honours — *Sep 2018 – Jun 2022*

Experience

Research Intern — Jun 2026 – Sep 2026
Chattermill | *GitHub* — *London, UK*

- Engineered an LLM-as-a-judge pipeline to evaluate text clusters, achieving a statistically significant positive correlation with human judgment
- Designed CLEAR, an LLM-guided clustering algorithm: beat k-means on 15/19 benchmarks ($p = 0.012$)
- Beat the best published SOTA result on 8 benchmarks with matched encoder and LLM
- Built a reproducible pipeline over 69K documents (≈1.3 cents per 100 docs); released as CLI tools

Research Intern — Jan 2025 – May 2025
LemonAI — *London, UK*

- Researched linguistic noise in large transcription datasets and its impact on NLP model robustness
- Architected hybrid heuristic and LLM-driven (Claude API) data augmentation pipeline to inject realistic adversarial noise, optimising data augmentation for out-of-distribution (OOD) generalisation

Projects

Cross-Lingual Factual Recall via Consistency-Driven RL | arXiv | GitHub — Jan 2026 – Jun 2026

- Investigated cross-lingual factual recall asymmetry in multilingual LLMs to identify latent bottlenecks inhibiting zero-shot knowledge transfer
- Approach: continual pretraining vs. SFT vs. consistency-driven RL (GRPO) to close this gap in 7B LLMs
- Second author; under review at ACL 2027
- Led the interpretability analysis: GRPO shifts factual processing to deeper, language-general layers

End-to-End Mobile Eye-Tracking with Deep Learning | *Personal Project* | GitHub — Jul 2025 – Nov 2025

- Built a data pipeline turning 515K GazeCapture frames into 1.5M+ augmented face/eye crops
- Designed a three-branch CNN reaching 1.4 cm test MAE
- Served it in a real-time mobile web app via TensorFlow Lite, wrote a technical article on Medium

Real-Time Facial Emotion Recognition via Transfer Learning | GitHub — Nov 2022 – Dec 2022

- Fine-tuned DenseNet169 on 40K+ faces: 81% accuracy vs. 71% for a custom CNN
- Led a team of 3; presented the real-time app at Le Wagon Demo Day

Skills

Specialisations: Machine Learning, Data Science
Statistics: Hypothesis Testing, Experimental Design
Programming Languages: Python, SQL, LaTeX
Libraries: pandas, NumPy, Matplotlib, PyTorch
Frameworks & Tools: Jupyter, Git, PostgreSQL, OpenAI API
Languages: Dutch (native), French (fluent), English (native)